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Analysis of ameliorative effects of 24-EBL on growth characteristics, photosynthetic pigment cascade and metal uptake of 60 day old plants of *B. juncea* plants under Cu metal stress

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Abstract : The present study describes the effect of 24-Epibrassinolide on growth characteristics, photosynthetic pigments content, elemental analysis and heavy metal uptake on plants of *Brassica juncea* L. under copper metal stress. The study revealed that there was an improvement in growth characteristics, pigments content and metal uptake under the influence of 24-Epibrassinolide treatments. In addition, 24-Epibrassinolide blocked copper metal uptake and accumulation in plants to a large extent.

Key Words: 24-Epibrassinolide, Heavy metal stress, growth characteristics, Photosynthetic pigments, metal uptake

INTRODUCTION

Global industrialization and human population explosion in the 21st century have led to contamination of air, soil and water by heavy metal contaminants. These contaminants have caused fatal effects on humans and the stability of the ecosystem [1]. Heavy metals are a category of elements that are non-biodegradable or partially degradable by microbial or chemical processes and tend to accumulate in the soil [2-3]. Copper (Cu) is an essential nutrient for plant functioning with a very low requirement and a corresponding elevated levels lead to toxicity [4-7]. The excessive amounts of ionic Cu in soil is phytotoxic and generally results in stunted root growth and shoot development, accompanied by leaf chlorosis [8-10].

BRs have a significant role in plants which are exposed to various biotic and abiotic stresses including drought, heavy metal, pesticide, infections, salt and viruses [11-12]. Certain plants are hyperaccumulator of heavy metals such as members of Brassicaceae family including *Brassica napus*, *Brassica juncea*, *Brassica monnieri*, *Brassica oleracea* etc. *Brassica juncea* L. is fast growing plant which has high biomass and grows well over heavily polluted soils [13]. In the present study, *Brassica juncea* plant were exposed to Cu stress and stress alleviation properties of BRs were studied by analysing the growth characteristics, photosynthetic pigment content and metal uptake.

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MATERIAL AND METHODS

Study material and treatments

Seeds were procured from seed technology division, Punjab Agriculture University, Ludhiana. A pot experiment was conducted in Botanical Garden of Guru Nanak Dev University, Amritsar. The seeds were surface sterilized with 0.01 % $HgCl_2$ for 2 min followed by rinsing 5 times with distilled water. The soil was added to pot containing, Clay: Sand: Manure in ratio of 3:1:1. The soil in the pots was given treatment of different concentration

of Cu metal solutions (0.25 mM Cu, 0.50 mM Cu and 0.75 mM Cu). Seeds were then soaked for 8 hrs, before sowing, in different concentration of 24-Epibrassinolide (0, 10-11, 10-9 and 10-7). Plants were then allowed to grow under natural conditions and harvested on 60th day.

Growth characteristics

Dry matter content and water content of plant sample was determined by method adapted from American standards of testing materials [14]. Plant samples were dried in a hot air oven at 102-105°C and the dried plant tissue was grounded into powder. The weight of the evaporating dish was measured and samples were then kept on the dish, followed re-weighing of the evaporating plate. The evaporating plates were then kept in the oven for 16 hours. After 16 hours the samples are cooled. The weight of the dried plant material and the evaporating dish was measured again. Relative water content was estimated by taking 0.5g of plant sample and saturated it in 100ml distilled water for 24 hr at 4oC in dark and then their turgid weights were recorded. This was then followed by oven dring at 65oC for 48 hr and their weights were recorded [15]. Metal tolerance index was measured as a quotient of dry weight of plant grown under copper treated and control plants [16].

Photosynthetic pigments content

Chlorophyll and carotenoid content was estimated by homogenizing 1g fresh plant tissue with 4ml of 80 % acetone followed by centrifugation using Eltek cooling centrifuge for 20 minutes at 13000 rpm at 4oC [17]. The supernatant was collected and absorbance of the supernantant was measured at 645 and 663 nm. Carotenoids were calculated by taking absorbance at 483 and 510 nm.

Anthocyanin content was determined by taking 1gm of fresh plant tissue and homogenized it using 3ml of extraction mixture i.e. acidified methanol (methanol: water: HCl, 79:20:1). The abosorbance of the supernatant was recorded at 530nm and 657nm [18].

Xanthophyll content was estimated by AOAC method according to which, 0.05g of dried plant material was transferred to 100ml flask. Then 30ml of extractant was transferred into the flask and shook well for 10-15 minutes. This was followed by addition of 2ml of 40 % methanolic KOH. The flask was then refluxed in water bath at 56oC and then kept in dark for 1 hour. Transfer 30ml of hexane in the flask and shook for 1 minute. The total volume was made upto 50ml with 10 % sodium sulphate solution. The samples were again incubated in dark for 1 hour. The upper phase was collected in 50 ml volumetric flask and the volume was made upto 50ml by hexane and measured at 474 nm [19].

Flavonoid content was determined by taking 1g of fresh plant tissue and homogenizing it in 3ml of absolute methanol and centrifuged for 20 minutes at 13000 rpm at 4oC. To 1ml of the plant extract 4ml of double distilled water was added. To this reaction mixture add 0.3ml of sodium nitrite ($NaNO_2$) and aluminum chloride ($AlCl_3$) followed by incubation for 5 minutes. Next 2ml sodium hydroxide (NaOH) was added and pink color was developed. Then 2.4 ml of distilled water was added to it and absorbance was recorded at 510nm [20].

Metal uptake

Determination of the copper uptake in plant samples was done by using Atomic Absorption Spectrophotometer (Shimadzu 6200). 0.5 g of dried plant samples were taken in digestion tubes. Sulphuric acid, nitric acid and perchloric acid in the ratio of 1:2:1 were taken for digestion [21].

Statistical analysis

Results obtained by using different methods were statistically analyzed by calculating mean, standard deviation, standard error and one way analysis of variance (ANOVA).

RESULTS AND DISCUSSION

Growth characteristic

Table1. Summarizes the results of effect of copper metal and 24-Epibrassinolide treatments on growth characteristics including dry matter content, water content, relative water content and metal tolerance index on 60 days old plants of *Brassica juncea* L.

Dry matter content was increased in response to plants treated with 24-EBL when compared to untreated plants. A significant increase was observed at 10-9 M EBL + 0.75 mM Cu treatment as compared to only 0.25 mM Cu treated plants. Water content was decreased in response to plants treated with 24-EBL when compared to untreated plants. A significant increase in the water content was found to be at 10-7 M EBL + 0.75mM Cu. Relative water

content was increased in response to plants treated with 24-EBL when compared to untreated plants. Maximum increase in the relative water content was found at 10-9 M EBL+ 0.25mM Cu treatment as compared to only 0.25mM Cu treated plants. Copper Tolerance Index was increased in response to plants treated with 24-EBL when compared to untreated plants. Maximum increase in the Tolerance Index was found to be at 0.25mM Cu + 10-7 M EBL treatment as compare to only 25mM Cu treated plants.

The growth characteristics in 60 days old *Brassica juncea* plants are generally observed to decrease due to copper metal treatment, particularily at highest Cu concentration i.e. 0.75 mM Cu. In contrast, application of BRs led to increase in dry matter, relative water and metal tolerance index.

Similar results were obtained by in a study conducted on *Cajanus cajan* L. (two cultivars, LRG 30 and ICPL 87) 6 day old seedlings grown under heavy metal stress [22]. Reduction in the dry matter content of plant is due to degradation of present leaves and new roots in response to depletion of moisture content [23]. Water content of *Brassica juncea* plants was increased in response to Cu metal stress whereas application of 24-EBL further enhanced the water content of the plants. Many reports suggest elevation in the water content in plants under heavy metal stress in response to application of BRs [24]. Relative water content was decreased with Cu treatment in comparison to control. Similarly, in rice seedlings under copper metal stress, the RWC of plant decreased with increase in the concentration of the metal [25]. Metal Tolerance Index of *Brassica juncea* plants under Cu metal stress was observed to decrease in response to increase in Cu concentration, whereas application of 24-EBL increased Metal Tolerance Index of plants under heavy metal stress. Similar observation of reduction in MTI in response to Zn and Cu stress was observed in wheat and rice plants [26].

Photosynthetic pigment content and metal uptake

Table2 and 3 summarize the results of effect of copper metal and 24-Epibrassinolide treatments on photosynthetic pigments content including chlorophyll, carotenoids, anthocyanin, xanthophylls and flavonoids and copper metal uptake on 60 days old plants of *Brassica juncea* L. Total Chlorophyll Content was increased in response to plants treated with 24-EBL when compared to untreated plants. Maximum increase in the total chlorophyll content was found at 0.75mM Cu + 10-9 M EBL treatment in comparison to only 0.75mM Cu treated plant. Total Carotenoid content was increased in response to plants treated with 24-EBL when compared to untreated plants. Anthocyanin content was increased in response to plants treated with 24-EBL when compared to untreated plants. Significant increase in the anthocyanin content was noticed at 0.25 mM Cu + 10-7 M EBL from 0.087 µg/g of FW to 0.106 µg/g of FW. Xanthophylls content was increased in response to plants treated with 24-EBL when compared to untreated plants. 10-7 M alone enhanced the xanthophylls content to the maximum (1.840 mg/g) as compared to 10-11 M EBL.

Total Flavonoid content was increased in response to plants treated with 24-EBL and same results were observed in Cu treated plants. Maximum increase in the flavonoid content at 0.25 mM Cu + 10-9 M EBL treatment in comparison to 0.25 mM Cu alone treated plants. Copper metal uptake was decreased in response to plants treated with 24-EBL when compared to untreated plants. With metal treatments, either alone or in combination with 24-EBL, maximum decrease in the copper metal uptake was noticed in 0.25 mM Cu + 10-7 M EBL.

Present work showed that Cu metal stress has an adverse effect on photosynthetic pigment cascade of plants. Decline in chlorophyll content was observed with increase in concentration of the copper metal [27]. The decline in chlorophyll content is attributed to interference of heavy metal in chlorophyll synthesis. The adequate supply of Fe and Mg ions is necessary for proper synthesis of chlorophyll pigment which is reported to be affected by heavy metal stress [28]. Indian mustard plants showed a decline in carotenoid content in response to Cu metal atress when compared to control plants. In the EBL treated plants, there was an observed increase in carotenoid content. Similar reports of decline in carotenoid and chlorophyll content were observed in Brassica juncea plants under Cd and Pb stress [29]. Several reports of elevation in levels of anthocyanin pigment in response to heavy metal stresses reveal their role in protection of plants under

variable stress condition [30]. Many reports suggest role of anthocyanin in quenching free radicals produced in plants under heavy metal stress. Activation of xanthophylls cycle and negative effect on chlorophyll and carotenoid content in *B. juncea* plants under cadmium stress suggest important role of xanthophylls in protecting photosynthetic machinery of plants [31]. Flavonoids are considered one of the most important secondary metabolite involved stress management in plants, by affecting its antioxidant capacity [32]. *Ginkgo biloba* plant under heavy metal stress demonstrated an increase in flavonoid content in response to Cu metal stress [20]. Similar observation has been made in *Ononis arvensis* plants under $CuSO_4$ stress [33].

Heavy metal uptake of Cu metal was observed to decrease with the application of 24-EBL as compared to the control plants. Similar finding were observed where Cu metal uptake and accumulation was observed to increase with increase in copper content in Chinese cabbage plant [34]. Another report, suggests increase in copper metal uptake and toxicity to plant in response to copper metal in *Brassica campestris* plant [35]. Ameliorative effect of 24-EBL on seedling growth, metal uptake and accumulation were determined in *B. juncea* plants under heavy metal stress. Metal uptake and accumulation was observed to be blocked in response to EBL treatments [36-37].

Table1: The effect of copper metal and 24-Epibrassinolide treatments on growth characteristics of 60 day old plant of *Brassica juncea* L.

Treatment	Dry Matter Content (Mean ± S.E) %	Water Content (Mean ± S.E) %	Relative Water Content (Mean ± S.E) %	Metal Tolerance Index (Mean ± S.E) %
0 (Control)	10.1 ± 0.25	89.1 ± 0.25	88.4 ± 16.94	100 ± 0
10^{-11} M EBL	11.1± 0.60	83.9 ± 0.60	78.6 ± 4.26	104.7 ± 3.16
10^{-9} M EBL	15.2 ± 1.55	86.5± 1.59	82.6 ± 5.24	136.8 ±10.7
10^{-7} M EBL	11.2 ± 1.99	88.6 ± 0.34	62.3 ± 3.49	100.4 ± 6.53
	F-Ratio $_{(df\ 3,8)}$= 1.225	F-Ratio $_{(df\ 3,8)}$= 1.225	F-Ratio $_{(df\ 3,8)}$= 1.484	F-Ratio $_{(df\ 3,8)}$= 7.209 *
0.25 mM Cu	9.6 ± 0.15	90.4 ± 0.15	64.5 ± 2.93	95.01 ± 0.84
10^{-11} M EBL + 0.25 mM Cu	10.33 ± 0.34	89.7 ± 0.34	69.3 ± 3.49	102.3 ± 0.76
10^{-9} M EBL+ 0.25 mM Cu	14.2 ± 3.05	84.8 ± 3.05	79.7 ± 4.51	121.9 ± 4.6
10^{-7} M EBL + 0.25 mM Cu	16.7 ± 3.36	83.1 ± 3.50	66.7 ± 3.17	132.8 ± 4.1
	F-Ratio $_{(df\ 3,8)}$= 2.378	F-Ratio $_{(df\ 3,8)}$=2.378	F-Ratio $_{(df\ 3,8)}$= 3.503	**F-Ratio $_{(df\ 3,8)}$= 29.754***
0.50 mM Cu	9.233 ± 0.22	90.8 ± 0.22	49.02 ± 1.83	91.5 ± 1.7
10^{-11} M EBL + 0.50 mM Cu	9.6 ± 0.31	90.4 ± 0.31	54.5 ± 2.31	95.01 ± 0.68
10^{-9} M EBL+ 0.50 mM Cu	11.5 ± 1.12	88.5 ± 1.12	53.2 ± 2.47	112.8 ± 8.8
10^{-7} M EBL + 0.50 mM Cu	9 ± 1.95	91 ± 1.95	54.2 ± 2.14	92.07 ± 15.4
	F-Ratio $_{(df\ 3,8)}$= 0.998	F-Ratio $_{(df\ 3,8)}$= 0.998	F-Ratio $_{(df\ 3,8)}$= 2.075	F-Ratio $_{(df\ 3,8)}$= 1.278
0.75 mM Cu	8.467 ± 0.37	91.5 ± 0.37	91.5 ± 0.37	83.8 ± 1.56
10^{-11} M EBL + 0.75 mM Cu	10.1 ± 0.252	89.9 ± 0.25	50.6 ± 0.41	96.21± 3.36
10^{-9} M EBL+ 0.75 mM Cu	12.5 ± 0.06	87.5 ± 0.05	68.6 ± 3.17	119.7 ± 2.64
10^{-7} M EBL + 0.75 mM Cu	9.5 ± 0.06	90.5 ± 0.05	75.8 ± 3.72	105.5 ± 8.56
	F-Ratio $_{(df\ 3,8)}$=56.348 *	F-Ratio $_{(df\ 3,8)}$= 56.347 *	F-Ratio$_{(df\ 3,8)}$= 16.866*	F-Ratio $_{(df\ 3,8)}$= 9.799 *

Table2: The effect of copper metal and 24-Epibrassinolide treatments on Chlorophyll, carotenoid and anthocyanin content of 60 day old plant of *Brassica juncea* L.

Treatment	Total Chlorophyll Content (Mean ± S.E) mg/g F.W.	Total Carotenoid Content (Mean ± S.E) mg/g F.W.	Anthocyanin Content (Mean ± S.E) µg/g of FW
0 (Control)	0.717 ± 0.12	0.239 ± 0.02	0.066 ± 0.010
10^{-11} M EBL	0.324 ± 0.05	0.179 ± 0.003	0.095 ± 0.002
10^{-9} M EBL	0.458 ± 0.06	0.198 ± 0.01	0.099 ± 0.003
10^{-7} M EBL	0.491 ± 0.04	0.187 ± 0.01	0.102 ±0.004
	F-Ratio $_{(df\,3,8)}$= 4.875 *	F-Ratio $_{(df\,3,8)}$= 4.022	F-Ratio $_{(df\,3,8)}$= 9.469 *
0.25 mM Cu	0.423 ± 0.05	0.144 ± 0.02	0.072 ± 0.010
10^{-11} M EBL + 0.25 mM Cu	0.429 ± 0.04	0.145 ± 0.02	0.076 ± 0.011
10^{-9} M EBL+ 0.25 mM Cu	0.558 ± 0.02	0.184 ± 0.02	0.084 ± 0.010
10^{-7} M EBL + 0.25 mM Cu	0.423 ± 0.13	0.160 ± 0.02	0.086 ± 0.008
	F-Ratio $_{(df\,3,8)}$= 1.148	F-Ratio $_{(df\,3,8)}$= 1.140	F-Ratio $_{(df\,3,8)}$= 0.414
0.50 mM Cu	0.312 ± 0.04	0.057 ± 0.05	0.083 ± 0.007
10^{-11} M EBL + 0.50 mM Cu	0.357 ± 0.03	0.125 ±0.02	0.086 ± 0.007
10^{-9} M EBL+ 0.50 mM Cu	0.480 ± 0.03	0.142 ±0.03	0.093 ± 0.006
10^{-7} M EBL + 0.50 mM Cu	0.476 ± 0.12	0.137 ±0.02	0.094 ± 0.007
	F-Ratio $_{(df\,3,8)}$= 1.545	F-Ratio $_{(df\,3,8)}$= 1.565	F-Ratio $_{(df\,3,8)}$= 0.669
0.75 mM Cu	0.267 ± 0.03	0.101 ± 0.03	0.087 ± 0.006
10^{-11} M EBL + 0.75 mM Cu	0.370 ± 0.02	0.212 ± 0.02	0.096 ± 0.005
10^{-9} M EBL+ 0.75 mM Cu	0.690 ± 0.05	0.215 ± 0.03	0.106 ± 0.010
10^{-7} M EBL + 0.75 mM Cu	0.593 ± 0.01	0.194 ± 0.03	0.105 ± 0.005
	F-Ratio $_{(df\,3,8)}$= 39.143 *	F-Ratio $_{(df\,3,8)}$= 4.319	F-Ratio $_{(df\,3,8)}$= 2.755

Table3: The effect of copper metal and 24-Epibrassinolide treatments xanthophylls, flavonoid and copper metal uptake of 60 day old plant of *Brassica juncea* L.

Treatment	Xanthophylls Content (Mean ± S.E) mg/g of DW	Total Flavonoid Content (Mean ± S.E) mg/g F.W.	Copper metal uptake (Mean ± S.E) µg/g D.W.
0 (Control)	0.12 ± 0.17	0.342 ± 0.09	28.4 ± 2.62
10^{-11} M EBL	1.374 ± 0.27	0.522 ± 0.02	39.3 ± 5.31
10^{-9} M EBL	1.640 ± 0.20	0.617 ± 0.02	33.0 ± 5.23
10^{-7} M EBL	1.840 ± 0.08	0.533 ± 0.008	33.0 ± 5.70
	F-Ratio $_{(df\,3,8)}$= 2.529	F-Ratio $_{(df\,3,8)}$= 5.664 *	F-Ratio $_{(df\,3,8)}$= 0.855 *
0.25 mM Cu	1.176 ± 0.05	0.392 ± 0.08	34.08 ± 4.32
10^{-11} M EBL + 0.25 mM Cu	1.412 ± 0.14	0.654 ± 0.03	18.7 ± 2.20
10^{-9} M EBL+ 0.25 mM Cu	1.593 ± 0.54	0.690 ± 0.01	12.0 ± 2.95
10^{-7} M EBL + 0.25 mM Cu	1.741 ± 0.05	0.539 ± 0.02	9.15 ± 4.58
	F-Ratio $_{(df\,3,8)}$= 1.134	**F-Ratio $_{(df\,3,8)}$= 8.893 ***	**F-Ratio $_{(df\,3,8)}$= 8.159 ***
0.50 mM Cu	1.130 ± 0.14	0.512 ± 0.04	37.82 ± 5.21
10^{-11} M EBL + 0.50 mM Cu	1.179 ± 0.13	0.580 ± 0.01	28.32 ± 5.84
10^{-9} M EBL+ 0.50 mM Cu	1.734 ± 0.53	0.697 ± 0.06	19.09 ± 4.79
10^{-7} M EBL + 0.50 mM Cu	1.468 ± 0.06	0.555 ± 0.10	13.79 ± 5.45
	F-Ratio $_{(df\,3,8)}$= 0.584	F-Ratio $_{(df\,3,8)}$= 1.685	F-Ratio $_{(df\,3,8)}$= 3.827*
0.75 mM Cu	1.082 ± 0.12	0.581 ± 0.02	60.04 ± 10.71
10^{-11} M EBL + 0.75 mM Cu	1.085 ± 0.26	0.625 ± 0.01	53.87 ± 4.28
10^{-9} M EBL+ 0.75 mM Cu	0.894 ± 0.04	0.638 ± 0.004	38.17± 4.48
10^{-7} M EBL + 0.75 mM Cu	1.412 ± 0.14	0.619 ± 0.03	31.69 ± 2.97
	F-Ratio $_{(df\,3,8)}$= 0.585	F-Ratio $_{(df\,3,8)}$= 1.359	**F-Ratio $_{(df\,3,8)}$= 4.323 ***

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